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Chapter 24

Growing Bankrolls: The Kelly Criterion and Rational Game Selection

Even though this is a book on the mathematics of poker, we will take a short break from direct poker examples to consider a topic that is of critical importance in many fields. This is the so-called "Kelly criterion," which is of importance in stock portfolio theory, blackjack, sports betting, and any other field where the goal is to take a starting bankroll and some positive-EV wager, the player decides what size wager to make, and hopes to grow the bankroll as fast as possible.

We made reference in Part I to the concept of ***utility,*** which is a relatively broad concept that has many facets beyond the scope of this book. For our purposes, utility reflects a person's preference for money in an adjusted manner. We ignore in this book concepts like "feeling charitable," even though these are important to a full understanding of utility.

As an example of this concept, suppose we are offered a choice of two outcomes. We may receive $20 immediately, or receive $10 immediately. Clearly, we prefer the $20. This is because we assign a higher utility to having $20 than having $10. Now consider the case where we can flip a fair coin for $1,000,000, or receive $500,000 immediately. In this case, we generally prefer the cash immediately. This is because the utility of a million dollars is not twice that of a half-million. 50% of the time we get the utility of a million, and 50% of the time we get the utility of zero.

We can construct a ***utility function*** which is a mapping of every bankroll x to its corresponding utility value $U(x)$. This function is unique to each individual and has the following properties:

- It is increasing for all values $x > 0$; that is, we never prefer a lower amount of money to the higher amount. If we did, if we had the higher amount we could simply give away the difference and increase our utility.
- It is usually concave downward. This simply means that the more money we have, the less utility an additional dollar will have. For example, the utility gained by a millionaire by finding a dollar in the street is much less than the utility gained by someone in poverty finding the same dollar.
- We generally treat it as though it were continuous (although since money is not strictly continuous, this is not precisely the case).

Suppose a bet B is proposed to us. There is some value of money X such that the utility $U(X)$ is equal to the utility of the bet, which is $<U(b)>$, where b represents the outcomes of B. For very small amounts of money, X might be extremely close to $<B>$, because utility is often close to linear for very small amounts. But on the other hand, if we consider the million dollar coin flip again, there is some value of money X which is the fair value of that bet to us in utility terms. This value X is called the ***certainty equivalent*** for ***B.***

What the Kelly criterion attempts to achieve is to utilize a utility function such that the growth rate of a bankroll or portfolio is maximized. The reason for this is that we can maximize overall EV by simply betting the whole bankroll on any +EV bet. However, when we lose, we have less bankroll to grow from the next trial. To accomplish this, we maximize the Kelly utility function $U(x)$ for each bet instead of maximizing dollar EV. For reasons that are beyond the scope of this book, the utility function that maximizes the growth rate per trial is:

$U(x) = \ln x$

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Interested readers can find a discussion of the derivation and selection of this utility function in any good finance textbook. Decisions that maximize this utility function are said to utilize the ***Kelly criterion*** or are simply called ***Kelly betting.*** Kelly betting has a number of important features that make it quite important in portfolio theory and in areas such as sports betting or blackjack. It:

- Maximizes the median bankroll after a given number of trials among all betting strategies.
- Minimizes the mean time required to reach a prespecified bankroll size among all betting strategies.
- Results in a distribution of bankrolls that is ***log-normal****;* that is, plotting the logarithm of the bankrolls results in a normal distribution. Because of the importance of this distribution in options trading and finance, it is quite well understood.

Example 24.1

As an example of how Kelly betting works, we consider the situation of a sports bettor. This bettor has some sort of model that allows him to pick games with a 3% edge; that is, his bets (including the bookmaker's juice) have a 3% advantage (they win 51.5% of the time and lose 48.5%), and are paid at even money. Now suppose this bettor has an initial bankroll of $5000, and can bet in any amounts. What are the correct amounts to bet that maximize the rate of growth of his bankroll?

To calculate this, we seek to maximize the expected value of the Kelly utility function. So if our bettor bets *x* dollars, then there are two possible outcomes. 51.5% of the time, the resultant bankroll will be (5000 + x), while 48.5% of the time, the resultant bankroll will be (5000 - *x).*

$<U(x)> = 0.515(\ln(5000 + x)) + 0.485(\ln(5000 - x))$.

To maximize this expression, we take the first derivative:

$0.515/(5000 + x) - 0.485/(5000 - x) = 0$
$x = 150$

So the amount that the bettor should bet on his 3% advantage is actually 3% of his bankroll; this is the case for all even-money bets. If he wins the lust bet, he then bets 3% of $5150, or $154.50, while if he loses, then he bets 3% of $4850, and so on. As this is not intended to be a text on gambling in general, we refer the reader to other sources for more information on this topic. The key concept we are trying to get across, which we will use in this section, is the substitution of the [ln x] utility function for the expected dollar value of a bankroll in making what we might call ***metagame decisions:*** those decisions which are not associated with a particular hand or a particular game, but rather, what games to play, what limits to play, or whether to play poker at all.

This ***log utility*** function leads us to a few comments about Kelly betting hi general that we should keep in mind:

- Kelly betting has a risk of rum of zero, assuming that we can subdivide our bankroll infinitely. The point is that when we lose, we scale our bet down immediately and when we win, we scale it up immediately.
- Kelly betting cannot apply directly to poker because there are not an infinite number of games at every limit. Although we might prefer it if our local cardrooms (or even online ones) would spread such limits as $1.02-$2.04 or $12.46-$24.92 in order to facilitate our

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ability to maximize our bankroll growth, such a development seems unlikely.

- The logarithmic nature of Kelly betting implies some interesting properties. For example, doubling our bankroll is worth as much to the positive as losing half of our bankroll is worth to the negative. And as our bankroll becomes much smaller, it becomes a utility disaster to lose fractions of it, while winning large absolute amounts from a large bankroll is worth comparably less.
- Kelly betting lends itself to considerably more risk than many players are used to from the risk of ruin model; this is because the rapidly-scaling nature of the strategy makes it more resistant to risk of ruin. By contrast, most players do not like to move back down after moving up in limits. If a player is willing to aggressively move up and down as his bankroll fluctuates, he can effectively play at different limits on a significantly shorter bankroll than it would take to play at the highest limit all the time.

With regard to this last, we also feel that typical players are often much more risk-averse than is really warranted with regard to opening poker businesses. For example, many players want to have just a 5% chance of losing their starting stake. Most entrepreneurs, however, would be ecstatic to learn that they have just a 5% chance of failing. On the other hand, it is rather unlikely that one will be able to build up a poker business and then sell it for a large amount (in the manner of many entrepreneurs). But we should be rational in assessing the chance that we will go broke versus the rewards available when we do not. Often cautious players spend far too much time playing at lower limits than is justified by their particular situation and forego large amounts of profit by doing so.

Kelly betting provides a risk-based framework for assessing the utility of growing a bankroll; we can apply this framework to the problem of choosing between games at different limits. We have a finite number of game options; our local cardrooms, whether brick and mortar establishments or online poker rooms, carry only specific limits, and a limited number of games. Or there are not players to fill up empty tables at the limits we would prefer to play. As a result, we are often faced with a decision between two games. In some cases, the decision is totally clear. If we can have higher expectation and less variance in one game, then choosing that game is clearly indicated. But this is often not the case, and we must decide between playing in a game with higher return and higher variance (in absolute dollar terms) and playing in a game with lower return but lower variance.

We can use Kelly utility as a guide to making this decision; given that we are unsure what game to play, and we are willing to move up and down between the two. As a result, we can investigate the degree to which the two games affect our ***utility*** function and decide which game to play for a particular bankroll size.

Before we go further with this model, however, let us consider a different angle on the Kelly utility; here the question is the ***optimal bankroll***. Previously we considered a fixed bankroll and found the optimal bet size; however, we can reverse the process for a game with a fixed bet and find the bankroll for which playing the game increases our Kelly utility maximally.

The Kelly utility for playing a game with a distribution of outcomes X is:

$$< \ln (B + x) > - \ln B$$

where B is the bankroll and x is a random variable selected from the distribution **X**.

Hopefully, this is fairly intuitive. Essentially, the change in the Kelly utility from playing the game is the expected value of the Kelly utility after playing the game minus the Kelly utility

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before playing it.

As a quick example, we return to a game we studied under risk of ruin. This is the game where we roll a die and win 1 unit when the die comes up 3-6, while losing 1 unit when the die comes up 1-2. This game effectively has a bet size of 1, since losing means a loss of 1 unit. If we have a bankroll B, then our Kelly utility from playing the game is:

$U(B) = (^1/_3)(\ln (B - 1)) + {}^2/_3 (\ln (B + 1)) - \ln B$

To maximize this, we take the derivative with respect to *B:*

$U'(B) = 1/(3(B - 1)) + 2/(3(B + 1)) - 1/B$
$0 = B(B + 1) + 2(B - 1)(B) - 3(B - 1)(B + 1)$
$0 = B^2 + B + 2B^2 - 2B - 3B^2 + 3$
$B = 3$

So for this game, the growth of Kelly utility from playing the game is maximized when the bankroll is equal to 3.

With this concept of optimal bankroll in mind, we can make the following assumptions for this model:

- We have two games to be compared; we will call these games game 1 and game *2.* Game 1 is the larger of the two games.
- We have known win rates μ_1 and μ_2 as well as standard deviations σ_1 and σ_2, for the two games. Both games have positive win rates and finite variance.
- We have a bankroll *B.*

Obviously, if we were forced to play one of these two games and we had a very large bankroll, we would play the higher of the two, while if our bankroll was quite small, we would play the lower. Our model here is that there is a bankroll cutoff c such that above *c* playing the higher game will increase our Kelly utility by more and below *c* the lower game will increase out-Kelly utility by more. We seek to find this bankroll cutoff.

For each game, we have some distribution of outcomes **X**. Our change in Kelly utility for each distribution is equal to:

$U = \langle \ln (B + x) \rangle - \ln B$
$U = \langle \ln B(1 + x/B) \rangle - \ln B$
$U = \langle \ln B \rangle + \langle \ln (1 + x/B) \rangle - \ln B$

$\langle \ln B \rangle$ is a constant, so we have:

$U = \langle \ln (1 + x/B) \rangle$

We can use the Taylor series expansion for ln (1 + t):

$$\ln(1 + t) = t - \frac{t^2}{2!} + \frac{t^3}{3!} - \frac{t^4}{4!} + [\ldots]$$

to find:

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$$\ln\left(1+\frac{x}{B}\right)=\frac{x}{B}-\frac{x^2}{2!\,B^2}+\frac{x^3}{3!\,B^3}-\frac{x^4}{4!\,B^4}+[\ldots]$$

If we use the assumption that B is substantially bigger than each outcome x, we can ignore the third and higher terms of this expansion, since they contain x^n in the numerator but $(n!)(B^n)$ in the denominator and thus approach zero extremely rapidly.

This leaves us with:

$$U \approx < \frac{x}{B}-\frac{x^2}{2B^2} >$$

For any distribution, we have $<x> = \mu$ and also the following:

$<(x-\mu)^2> = \sigma 2$
$<(x^2 - 2x\mu + \mu^2)> \sigma^2$
$<x^2> - 2 <x> \mu + \mu^2 = \sigma^2$
$<x^2> - \mu^2 = \sigma^2$
$<x^2> = \mu^2 + \sigma^2$

Substituting into our formula for *U*, we have:

$$U \approx \frac{\mu}{B}-\frac{\mu^2+\sigma^2}{2B^2}$$

The cutoff bankroll c that we are looking for is the bankroll at which the utility of Game 1 equals the utility of Game 2. Above this value, we will play Game 1; below it, we will play Game 2.

$$\frac{\mu_1}{c}-\frac{\mu_1^2+\sigma_1^2}{2c^2}=\frac{\mu_2}{c}-\frac{\mu_2^2+\sigma_2^2}{2c^2}$$

$2c\mu_1 - \mu_1^2 - \sigma_1^2 = 2c\mu_2 - \mu_2^2 - \sigma_2^2$
$2c(\mu_1 - \mu_2) = \mu_1^2 - \mu_2^2 + \sigma_1^2 - \sigma_2^2$
$2c = \mu_1 + \mu_2 + (\sigma_1^2 - \sigma_2^2)/(\mu_1 - \mu_2)$

Because in poker, μ_1 and μ_2 are normally extremely small compared to the variances, we can abbreviate this formula to:

$$c \approx \frac{\sigma_1^2-\sigma_2^2}{2(\mu_1-\mu_2)} \qquad \textbf{(24.1)}$$

Let's look at a more concrete example.

Example 24.2

Suppose there are two games we're considering playing in a cardroom; one is \$20-40 holdem, where we have a win rate of \$35/hr with a standard deviation of \$400/hr, and the other is \$40-80 holdem, where we have a win rate of \$60/hr, with a standard deviation of \$800/hr.

Then the cutoff bankroll is:

$2c = 60 + 35 + (800^2 - 400^2)/(60 - 35)$
$c = \$19{,}295$

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This means that with a bankroll of more than this amount, we should play the $40-80 game, while with less than this amount we should prefer to play $20-40 from a utility standpoint.

This framework provides a means for comparing the risk-adjusted utility of playing in two different games so that we can make the best decision about which game to play ill a quantitative manner. The primary limitation on using this formula is that it is invalid for distributions of outcomes where some of the outcomes are of comparable size to the bankroll (in these cases, dropping the later terms of the Taylor expansion is not reasonable). This occurs most frequently in tournaments, especially those with large fields, where the top payouts may exceed 500 buyins. Nevertheless, for all types of ring games we can quantify' the risk-adjusted value of playing in any two different games using this method.

Key Concepts

- When making real-world decisions, we cannot simply use expectation as a guide; instead we must consider utility. We can assign a utility function and evaluate outcomes in terms of utility.
- One common utility function which is studied very heavily in financial markets because of its special properties in portfolio theory is the Kelly criterion, or
- $U(x) = \ln x$.
- By substituting the Kelly criterion into two possible sets of game conditions, we can find the utility of playing in each game. When the games have appropriate conditions, we can identify a bankroll level above which we should play the larger game, and below which we should play the smaller.

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